SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO4: Formulate data-driven web solutions using XML, XSLT, and JSP within the MVC framework. (L4)

Assessment Tool: Pseudocode Writing Task (Algorithm Design)

Weightage: 20%

QUESTIONS: Total Marks: 25

1: XML Parsing Algorithm

Write a detailed **pseudocode algorithm** to parse an XML document that contains structured data such as user or product details. Your algorithm should clearly describe the steps involved in loading the XML file, accessing its elements, traversing through nodes, extracting required data, and displaying the output in a readable format. Ensure that the logic handles hierarchical structure properly.

```
ALGORITHM ParseXMLDocument
 INPUT: xml_file_path (String)
 OUTPUT: Formatted elements printed to console

 TRY
 Initialize xml_parser = Create new XMLParser()
 Initialize xml_doc = xml_parser.Load(xml_file_path)
 CATCH Exception
 PRINT "Error: Loading XML failed."
 RETURN
 END TRY
 Initialize root_node = xml_doc.GetRootElement()
 Call TraverseNodes(root_node, 0)
END ALGORITHM

PROCEDURE TraverseNodes(current_node, depth)
 Initialize indent = Repeat(" ", depth)
 IF current_node.HasAttributes() THEN
 FOR EACH attr IN current_node.GetAttributes()
 PRINT indent + "Attr: " + attr.GetName() + " = " + attr.GetValue()
 END FOR
 END IF
```

```
 IF current_node.HasChildren() THEN
 PRINT indent + "Node: " + current_node.GetName()
 FOR EACH child IN current_node.GetChildren()
 Call TraverseNodes(child, depth + 1)
 END FOR
 ELSE
 PRINT indent + current_node.GetName() + ": " + current_node.GetTextValue()  END IF
END PROCEDURE
```

2: MVC Workflow

Write a **pseudocode representation** of the MVC (Model–View–Controller) workflow in a web application. Your answer should clearly illustrate how a user request is received, processed by the Controller, how the Model interacts with the database, and how the View generates and returns the response to the user. Include all major steps involved in request handling and response generation.

```
ALGORITHM MVC_Workflow_Handler
 INPUT: user_http_request (HTTPRequest)
 OUTPUT: http_response (HTTPResponse)

 // Step 1: Controller intercepting the request
 Initialize controller = Router.RouteToController(user_http_request)
 Initialize action_parameters = controller.ExtractRequestParameters(user_http_request)

 // Step 2: Controller interacts with Model
 Initialize model_object = controller.SelectModel(action_parameters)
 TRY
 Initialize db_connection = DatabaseManager.GetConnection()
 Initialize query_result =
db_connection.ExecuteQuery(model_object.GetSQLQuery(action_parameters))
model_object.PopulateState(query_result)
 CATCH DatabaseException
 model_object.SetErrorState("DB_FAIL")
 FINALLY
 db_connection.Close()
 END TRY

 // Step 3: Controller passes Model state to View and returns Response
 Initialize view_template = controller.SelectView(action_parameters)
 Initialize rendered_html = view_template.Render(model_object.GetState())  Initialize
http_response = Create new HTTPResponse(rendered_html)
 RETURN http_response
END ALGORITHM
```

3: JSP Request Flow

Write a detailed **pseudocode algorithm** to explain the request–response flow in a JSP-based web application. Your algorithm should include steps such as client request initiation, server

processing, interaction with backend components (like Servlets or database), and generation of dynamic content that is sent back to the client.

```
ALGORITHM JSP_Request_Response_Flow
 INPUT: client_http_request (HTTPRequest)
 OUTPUT: client_http_response (HTTPResponse)

 // Step 1: Client Request Initiation
 Initialize jsp_name = client_http_request.GetPath()

 // Step 2: Translation and Compilation Check
 IF NOT ServletContainer.IsCompiled(jsp_name) OR ServletContainer.IsModified(jsp_name)
THEN
 Initialize servlet_java_source = JSPCompiler.TranslateToServletSource(jsp_name)  Initialize
servlet_class = JavaCompiler.Compile(servlet_java_source)
 ServletContainer.LoadServlet(servlet_class)
 END IF

 // Step 3: Execution and Backend Interaction
 Initialize servlet_instance = ServletContainer.GetServletInstance(jsp_name)  Initialize
jsp_context = Create new JSPContext(client_http_request)
 IF jsp_context.RequiresDatabaseAccess() THEN
 Initialize db_conn = ConnectionPool.GetConnection()
 jsp_context.SetAttribute("pageData", db_conn.FetchData(jsp_context.GetQueryString()))
db_conn.Close()
 END IF

 // Step 4: Content Generation and Response Return
 Initialize out_stream = ServletContainer.GetOutputStream(client_http_response)  Initialize
rendered_html = servlet_instance._jspService(jsp_context, out_stream)
client_http_response.SetBody(rendered_html)
 client_http_response.Send()
END ALGORITHM
```

4: Database Retrieval

Write a **pseudocode algorithm** to retrieve data from a database. Your answer should include steps for establishing a database connection, executing SQL queries, processing the result set, and displaying the retrieved data. Clearly represent the sequence of operations involved in database interaction.

```
ALGORITHM Database_Retrieval_Flow
 INPUT: db_credentials (DBConfig), query_string (String)
 OUTPUT: result_set_list (List of Objects)

 Initialize db_connection = NULL, prepared_statement = NULL, result_set = NULL  Initialize
result_set_list = Create empty List()
 TRY
 // Step 1: Establish Database Connection
 db_connection = DriverManager.GetConnection(db_credentials.GetURL(),
```

```
db_credentials.GetUsername(), db_credentials.GetPassword())
 // Step 2: Prepare SQL Query
 prepared_statement = db_connection.PrepareStatement(query_string)  // Step 3: Execute SQL Query
 result_set = prepared_statement.ExecuteQuery()
 // Step 4: Process the Result Set
 WHILE result_set.Next() DO
 Initialize record = Create new Map()
 FOR EACH column IN result_set.GetMetaData().GetColumns() DO
 record.Put(column.GetName(), result_set.GetObject(column.GetName()))  END FOR
 result_set_list.Add(record)
 END WHILE
 CATCH SQLException AS sql_error
 PRINT "Database error: " + sql_error.GetMessage()
 FINALLY
 // Step 5: Close Resources
 IF result_set != NULL THEN result_set.Close() END IF
 IF prepared_statement != NULL THEN prepared_statement.Close() END IF  IF db_connection != NULL THEN db_connection.Close() END IF
 END TRY
 RETURN result_set_list
END ALGORITHM
```

5: PHP Validation

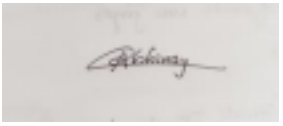
Write a **pseudocode algorithm** for validating user input in a web form using PHP concepts. The algorithm should include checks for empty fields, validation of input formats (such as email and password), handling invalid inputs with appropriate error messages, and allowing submission only when all validations are satisfied.

```
ALGORITHM PHP_User_Input_Validation
 INPUT: post_data (Array from $_POST)
 OUTPUT: validation_results (Map containing 'is_valid' [Boolean] and 'errors' [List of Strings])

 Initialize validation_results = Create new Map()
 Initialize errors = Create empty List()
 validation_results.Put("is_valid", TRUE)

 // Step 1: Extract and Sanitize Inputs
 Initialize email_input = TrimString(post_data.Get("email"))
 Initialize password_input = post_data.Get("password")
 // Step 2: Check for Empty Fields
 IF IsEmpty(email_input) THEN
 errors.Add("Email field cannot be empty.")
 END IF
```

```
IF IsEmpty(password_input) THEN
errors.Add("Password field cannot be empty.")
END IF

// Step 3: Validate Email Format
IF NOT IsEmpty(email_input) THEN
IF NOT IsValidEmailPattern(email_input) THEN
errors.Add("Invalid email format.")
END IF
END IF

// Step 4: Validate Password Strength
IF NOT IsEmpty(password_input) THEN
IF Length(password_input) < 8 OR NOT HasUppercase(password_input) OR NOT
HasNumber(password_input) THEN
errors.Add("Password must be at least 8 characters, contain one uppercase letter, and one
number.")
END IF
END IF

// Step 5: Handle Validation Results
IF Length(errors) > 0 THEN
validation_results.Put("is_valid", FALSE)
validation_results.Put("errors", errors)
ELSE
validation_results.Put("is_valid", TRUE)
END IF
RETURN validation_results
END ALGORITHM
```

Code of Conduct:

I ____G.Abhinay(192373062)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation

Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient